

Intelligent Collision Avoidance for Multi Agent Mobile Robots

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Abstract. This chapter presents a newly developed mobile robot based multi-agent system with capabilities of robust motion control and intelligent collision avoidance. The system consists of three mobile robots. One main robot acts as a master and the other two act as slaves. The master intelligently takes decisions as to which action to perform to avoid obstacles and collisions. The master mobile robot has the capability to swerve around a static or moving object when necessary. All possible conditions have been coordinated in a fuzzy knowledge base which is used to make a decision on the required maneuver to avoid a collision with a slave robot that the mobile robot may encounter on its driving lane. The proposed research has been carried out to simulate a real car driving regime on roads where the driver may not react properly. The system is implemented on a robot experimental test bench and some experimental results are presented and discussed.

Keywords: Fuzzy Logic Control (FLC), Multi-agent Mobile Robot, Collision Avoidance.

1 Introduction

Automobile collision avoidance systems have received wide attention by industry and academia over the last ten years, with the aim of reducing road accidents caused by driver error. The key issue in most collision avoidance systems is the decision making required to avoid collision. Most recent techniques utilized by collision avoidance systems depend on deterministic calculations for deciding the required action to safely avoid a collision and subsequent accident [1–3]. This paper instead investigates on an intelligent collision avoidance system that implements the decision-making process by Fuzzy Logic Control (FLC) [4–7]. This helps the system to consider several critical cases to avoid a collision, namely velocity reduction, lateral path variation (or swerving), safe stopping, and velocity change of the slave mobile robots. Communication between the master and slave robots is via an RF communication module [8] and [9].

The remainder of the chapter is organized as follows. Section 2 provides a brief description of the proposed multi-agent robot system. Section 3 discusses the

proposed control strategy. Section 4 describes the system wireless communication and data exchange. The high level control is explained in Section 5. The low level control system is exhibited in Section 6. Section 7 discusses the practical results conducted from a practical test. Conclusions are provided in Section 8.

Notation

R_{12} denotes the distance between robot R_1 and robot R_2

R_{13} denotes the distance between robot R_1 and robot R_2

V_1 denotes the speed of master robot R_1

V_3 denotes the speed of slave robot R_3

S denotes velocity membership function ‘Slow’

H denotes velocity membership function ‘High’

S denotes distance membership function ‘Small’

M denotes distance membership function ‘Medium’

B denotes distance membership function ‘Big’

VB denotes distance membership function ‘Very Big’

2 Proposed Multi Agent Robot System

Multi-agent systems have received wide research attention by academic and industry. This is due to their inherited system complexity in behavior and to the challenges for providing an intelligent control strategy, which meets all disciplines of multi-agent robot motion control while moving in a specific domain environment. A different application of multi-agent mobile robot systems, e.g., unmanned air craft and spaceflight systems, unmanned underwater vehicles, walking robots and automotive and transportation robot systems etc., require such a control strategy to enable the system achieve the desired goals which has been designed for [10–14].

For any multi-agent robot system applications there must be motion coordination, which aims to intelligently coordinate the motion of the agents in a specific working domain as well as arranging what is most needed to avoid collision between one or more agents while moving.

Fig. 1 shows a schematic of the proposed mobile robot system for robot-to-robot collision avoidance. The multi-agent mobile robot system consists of three robots. The master robot, R_1 , and slave, R_2 , move on a circuit path in one direction, and the second slave robot, R_3 , travels in the opposite direction on a parallel path.

A virtual trajectory is used to enable an S-shape overtaking maneuver by the master robot as an option to avoid collision. The S-shape maneuver allows the master robot to ‘swerve’ around the third-party mobile. The decision to perform an S-shape motion is based on an intelligent fuzzy inference engine, as